

Method 1:

1. Evaluate $\int \sin^3 x \cos^3 x dx$. $u = \sin x$
 $du = \cos x dx$

$$= \int \sin^2 x \cos^2 x \cos x dx$$

$$= \int \sin^2 x (1 - \sin^2 x) \cos x dx$$

$$= \int u^2 (1 - u^2) du$$

$$= \int (u^2 - u^4) du$$

$$= \frac{1}{3} u^3 - \frac{1}{5} u^5 + C$$

$$= \frac{1}{3} \sin^3 x - \frac{1}{5} \sin^5 x + C$$

2. Evaluate $\int \sec^3 x \tan x dx$.

$u = \sec x$
 $du = \sec x \tan x dx$

$$= \int \sec^2 x \sec x \tan x dx$$

$$= \int u^2 du$$

$$= \frac{1}{3} u^3 + C$$

$$= \frac{1}{3} \sec^3 x + C.$$

Method 2:

$u = \cos x$
 $du = -\sin x dx$

$$= \int \sin^2 x \cos^3 x \sin x dx$$

$$= \int (1 - \cos^2 x) \cos^3 x \sin x dx$$

$$= - \int (1 - u^2) u^3 du$$

$$= - \int (u^3 - u^5) du$$

$$= - \left(\frac{1}{4} u^4 - \frac{1}{6} u^6 \right) + C$$

$$= -\frac{1}{4} \cos^4 x + \frac{1}{6} \cos^6 x + C$$

Formula Sheet

$$\sin^2 x + \cos^2 x = 1, \quad \sin^2 x = 1 - \cos^2 x, \quad \cos^2 x = 1 - \sin^2 x$$

$$\sec^2 x = \tan^2 x + 1, \quad \tan^2 x = \sec^2 x - 1$$

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}, \quad \sin 2x = 2 \sin x \cos x$$

$$\int \cot x \csc x dx = -\csc x + C, \quad \int \csc^2 x dx = -\cot x + C$$

$$\int \tan x dx = \ln |\sec x| + C, \quad \int \cot x dx = \ln |\sin x| + C$$

$$\int \sec x dx = \ln |\sec x + \tan x| + C, \quad \int \csc x dx = \ln |\csc x - \cot x| + C$$